

Device Fabrication/ Fabrication Of Chips And Microprocessors:

The fabrication of a device involves a multi-step procedure for manufacturing and assembling numerous technical components. In semiconductor engineering, this procedure is applied to the production of chips, integrated circuits (ICs), and microprocessors, where carefully controlled physical and chemical processes are used to build highly compact and efficient electronic circuits on silicon wafers

Basic Steps in Chip Fabrication

Fabrication of chips is an incredibly precise process that turns raw silicon into a functional brain for an electronic device Steps for Fabrication of chips:

1. Silicon Wafer Preparation
2. Deposition & Coating
3. Photolithography
4. Etching & Ion Implantation

Q.Explain the steps in chip fabrication.

1. Silicon Wafer Preparation

The process begins with Electronic grade silicon. This silicon is melted and formed into a single large crystal called an ingot. Slicing and Polishing →The ingot is being sliced with a diamond saw into thin circular wafers. Then the wafers are polished to a mirror-like finish to ensure there are no microscopic bumps or scratches.



2. Deposition & Coating

To create the different layers of the circuit, different materials must be added to the silicon wafer.

- Insulation → A layer of silicon dioxide is grown on the surface using high-temperature oxidation.
- Deposition → Thin films of conducting metals or semiconducting materials are added by Chemical Vapour Deposition (CVD) or Physical Vapour Deposition (PVD).

3. Photolithography

This is the process where the actual circuit design is being transferred onto the wafer.

- Photoresist → The wafer is coated with a light-sensitive liquid called photoresist.

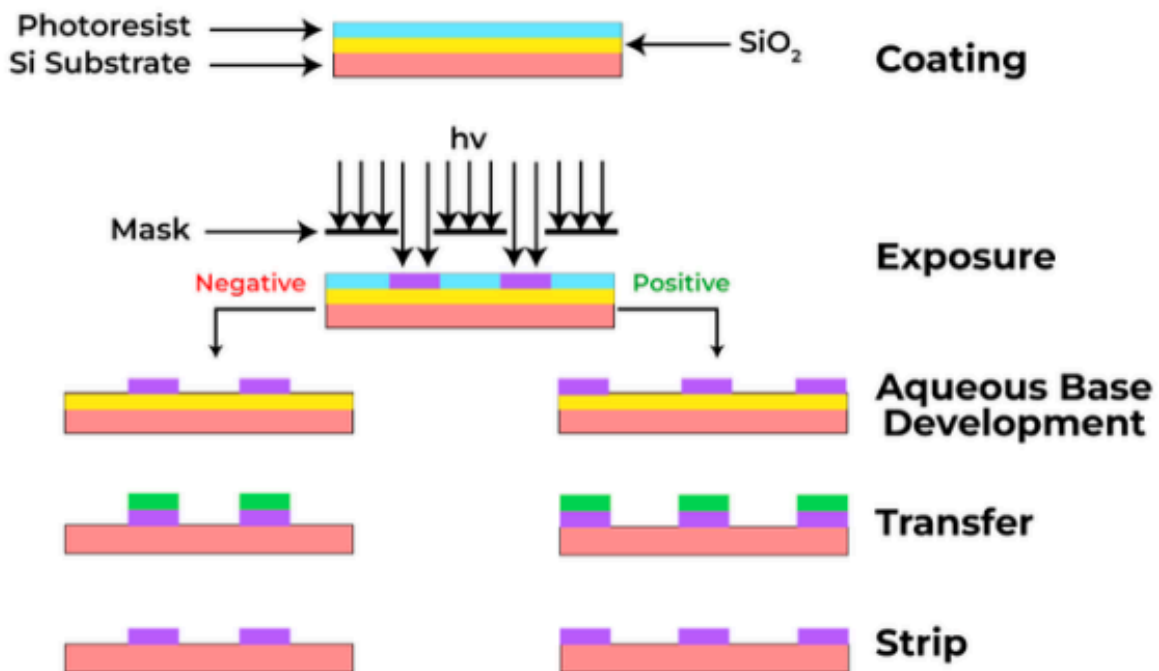
- Exposure → Deep Ultraviolet (DUV) or Extreme Ultraviolet (EUV) light is projected through a mask (a template of the circuit) onto the wafer.
- Development → The wafer is washed in a chemical developer. Depending on the type of photoresist used, the light-exposed areas are either hardened or washed away, leaving a precise pattern behind.

4. Etching & Ion Implantation

Now that the pattern is printed, the physical structure of the chip is carved out.

- Etching: → A dry gas or wet chemical removes the unprotected layers of material, leaving only the pattern defined by photolithography.
- Ion Implantation: → The wafer is bombarded with ions (like Boron or Phosphorus). These ions penetrate the silicon to change its electrical properties, creating the Source and Drain regions of the transistors.

Photolithographic Process



Q. What is Photolithography?

Photolithography is a technique that is used to define the shape of micromachined structures on a wafer.

Photolithography, also termed optical lithography or UV lithography, is a process used in micro-fabrication to pattern parts of a thin film or the bulk of a substrate. It uses light to transfer a geometric pattern from a photomask to a light-sensitive chemical "photoresist", or simply "resist," on the substrate. A series of chemical treatments then either engraves the exposure pattern into, or enables deposition of a new material in the desired pattern upon, the material

underneath the photo resist. For example, in complex integrated circuits, a modern CMOS wafer will go through the photolithographic cycle up to 50 times.

Processing Steps in Photo-Lithography:

- **Dehydration:** Removes the water molecules from the substrate surface, makes the surface hydrophobic, which increase the adhesion between substrate and photo-resist
- **Priming:** In this step, an adhesion promoter known as primer is added to the substrate surface.
- **Application of photo-resist:** Photo-resist is applied to substrate surface through either spin coating or spray coating
- **Soft Bake / Pre-bake:** Excess solvent in photo-resist is driven out of the film by heat
- **Pattern Alignment:** Pattern of mask is aligned with the pattern on substrate
- **Post-exposure Bake:** In this step, the photo-resist is further cross-linked by heating
- **Photo-resist Exposure:** Photo-resist undergoes cross-linking under the illumination of UV light. **Photo-resist Development:** Exposed positive resist or unexposed negative resist is dissolved in developer leaving needed part on substrate surface.
- **Hard Bake:** Photo-resist is further baked to remove pin-holes and make it denser to facilitate the later etching step.
- **Phot-Resist Removal:** Photo-resist is removed by chemicals that can either dissolve or oxidize photo-resist.

Q. Write a short note on photoresist used in photolithography.”/“Explain photoresist and its role in semiconductor fabrication.”

Photoresist is a light-sensitive polymer material used in photolithography to transfer circuit patterns onto a silicon wafer during semiconductor fabrication. It acts as a temporary protective layer that defines specific regions for etching or doping.

Composition of Photoresist

A typical photoresist consists of:

Polymer (Resin): Forms the structural base.

Photoactive Compound (PAC): Reacts chemically when exposed to light.

Solvent: Keeps the resist in liquid form for spin coating.

Additives: Improve adhesion and stability.

Requirement of Photoresist

High resolution

-Thinner PR film has higher the resolution

-Thinner PR film, the lower the etching and ion implantation resistance

High etch resistance

Good adhesion

-Wider process latitude Higher tolerance to process conditions like spin rate, baking temperature and exposure flux

Working Principle

- The wafer is coated with liquid photoresist.
- It is exposed to ultraviolet (UV) light through a mask.

- Light causes a chemical change in the exposed regions.
- A developer solution removes either the exposed or unexposed regions.
- A patterned resist layer remains on the wafer.
- This patterned layer protects selected areas during etching or ion implantation.

Positive and Negative Photoresist

There are two types of photoresist: **positive and negative**.

For **positive resists**, the resist is exposed with UV light wherever the underlying material is to be removed. In these resists, exposure to the UV light changes the chemical structure of the resist so that it becomes more soluble in the developer. The exposed resist is then washed away by the developer solution, leaving windows of the bare underlying material. In other words, "whatever shows, goes." The mask, therefore, contains an exact copy of the pattern which is to remain on the wafer.

Negative resists behave in just the opposite manner. Exposure to the UV light causes the negative resist to become polymerized, and more difficult to dissolve. Therefore, the negative resist remains on the surface wherever it is exposed, and the developer solution removes only the unexposed portions. Masks used for negative photoresists, therefore, contain the inverse (or photographic "negative") of the pattern to be transferred. The figure below shows the pattern differences generated from the use of positive and negative resist.